Features, Labels, Targets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach field roles through explicit timing evidence.

Supply full schemas and example rows, expose disguised outcomes and assess a new task with unknown future label.

Keep one habit: Ask when each value existed, which definition is preserved and what the feature-selection code does not establish.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary **target**: Separate **prediction input**s from observed **target**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach **feature**s, **label**s and **prediction time** in two 45-minute sessions using fictional records. Students must define the decision moment before judging columns.

The key distinction is that **label**s may be known for historical **training** but must not be fed into **prediction**s for new cases.

This lesson prepares **input**s; it does not demonstrate a trained model or establish that loan **prediction** is useful. Use paper records or local Python, without accounts or personal borrowing data.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

They are reserved to evaluate the fixed choice. Using their **label**s to choose it makes them part of development.

Printed lesson and workbook

Two cards **label**led checkout and later return

Optional Python 3 editor; no external packages

Blank field meaning unit availability role table

Opening retrieval

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish **fitting** a rule from applying a fixed rule

Reference example, not the only acceptable wording: The **input** is width x and the **label** is the observed category 0 or 1. Candidates are 2, 4 and 6.

Selection uses mistakes on the four **training** cards only; held out **label**s must not guide it.

End goal

By the end, I can explain **feature**s, **label**s, **target**s using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

The ML Lifecycle

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.
</details>

What the learner must demonstrate

A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

Concept to explain

Features are the information available at **prediction time**; the **target** is the value the model should estimate; **label**s are observed **target** values in examples.

Term	Meaning
feature	input variable used for prediction
target	quantity or category to estimate
label	observed target value in a dataset
row	one example with fields

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach availability from meaning and timestamps

> Separate the library, plant and puzzle target definitions.

Read the library task and source snapshot

Week 1 separated **training** examples from testing examples. This week asks a different question: could each **input** actually be known when a **prediction** is needed? A historical table may contain facts added after the event.

A model trained with those facts can appear successful even though the **input**s will be missing when it is used.

Use fictional library loans. At the instant a book is checked out, predict whether it will be returned more than 14 days later.

For completed historical loans, **label** 1 means days_to_return is greater than 14; **label** 0 means it is at most 14. Exactly 14 days is on time. Assume no changed due dates in this exercise.

All records and values are invented; do not collect classmates' borrowing histories.

queue_at_checkout is a saved count of waiting requests at that instant. It is not the queue when we later export the table. Candidate means available to consider, not proven useful or appropriate.

Exclude the arbitrary loan identifier from this exercise; keep it separately to track records.

Library feature contract	Declared rule
Prediction time	Checkout, before the return outcome
Feature order	pages then queue_at_checkout; page count then saved waiting-request count
Values	Positive integer pages; nonnegative integer queue, including 0
Target	returned_late=1 when days_to_return>14, else 0 for completed loans; exactly 14 is on time
Tracking metadata	Arbitrary loan_id is excluded from predictors in this exercise
Excluded inputs	days_to_return, returned_late, fee_after_return and post-checkout renewals

Library feature contract	Declared rule
Implementation limit	Feature selection code assumes valid fields and does not fit a model, validate arbitrary records or predict a new outcome

Trace fixed-order features and historical label

The **feature** order is always pages first, queue count second. The first **output** is [120, 2]; the separate **label** is 1 because this historical loan took 18 days.

The new loan produces [80, 0] even though its return outcome does not yet exist. The same **feature** function serves past and new records.

It deliberately selects allowed fields rather than copying every column except returned_late.

This code prepares **input**s; it does not fit a model or predict lateness. It assumes the two **feature** fields contain valid nonnegative integer values and pages is positive. A missing required field raises KeyError.

A real pipeline needs explicit validation and a missing-data policy.

Read the separate plant-capture task

Choose **feature**s that exist when a plant **prediction** must be made.

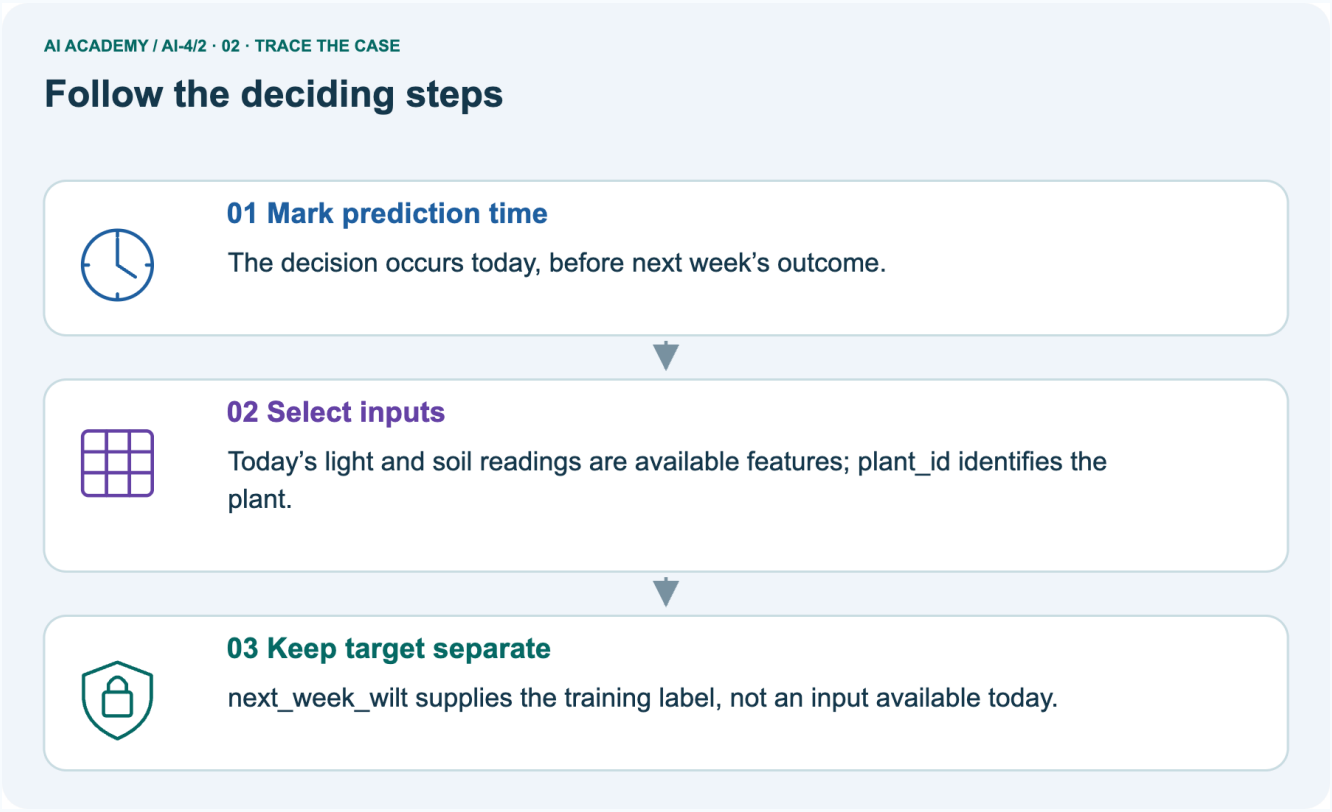
Predict next-week wilting at today's photo capture. Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Separate task contracts. Library lateness is predicted at checkout using pages/queue snapshot. The pictured plant task predicts next-week wilting at today's capture using today_light and today_soil_moisture; plant_id is metadata and next_week_wilt is a later target. The added tomorrow_soil_moisture field is explicitly an observed reading captured tomorrow, not a forecast available today.

Reject it for today's inputs even if a later export contains it. These are fictional records, not real plant or borrowing data.

Trace present features and future target



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Mark prediction time	The decision occurs today, before next week's outcome.
2. Select inputs	Today's light and soil readings are available features; plant_id identifies the plant.
3. Keep target separate	next_week_wilt supplies the training label, not an input available today.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict adding observed tomorrow data

Change: A table adds tomorrow_soil_moisture as an **input**.

Prediction: Reject it for this **prediction**-time contract.

Reason: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

State candidate-usefulness and causal limits

Eligible fields are not automatically useful or causal. Measure their predictive value with a valid split.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Reference · Generic field-timing questions

For every proposed **input**, ask what it means, when it was recorded, and whether the same definition can be supplied at checkout. A field in the historical export is not automatically available at **prediction time**.

An **input** calculated from later information is also unavailable, even if its name does not mention the outcome.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Mark **feature**s and **target**s in three datasets and reject information unavailable at real **prediction time**.

MODEL IDEA **Feature**s are the information available at **prediction time**; the **target** is the value the model should estimate; **label**s are observed **target** values in examples.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Plant row: leaf_width=6 cm, color_score=0.72, health=spotted. **Feature**s are leaf_width and color_score because they exist before **prediction**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Plant row: leaf_width=6 cm, color_score=0.72, health=spotted.

STEP 2 **Feature**s are leaf_width and color_score because they exist before **prediction**.

STEP 3 **Target** is health; the stored value spotted is this row's **label**.

STEP 4 A treatment-result field recorded later is not a valid **prediction**-time **feature**.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Rows contain leaf length, edge type and healthy/spotted **label**.

Identify **feature**s, **target** and **prediction**-time **input**s.

Exact explanation to model

Length and edge type are candidate **feature**s; the **label** is the **target**. Do not supply the true **target** as an **input** to predicting it. Evaluate **feature** usefulness.

Explain the added worked example

Key terms

Term	Meaning
Feature contract	A specification of input meaning units order and timing
Target	The outcome a prediction aims to estimate
Prediction time	The moment when the prediction must be made
Target leakage	Predictor information that reveals the outcome but is unavailable at prediction time

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

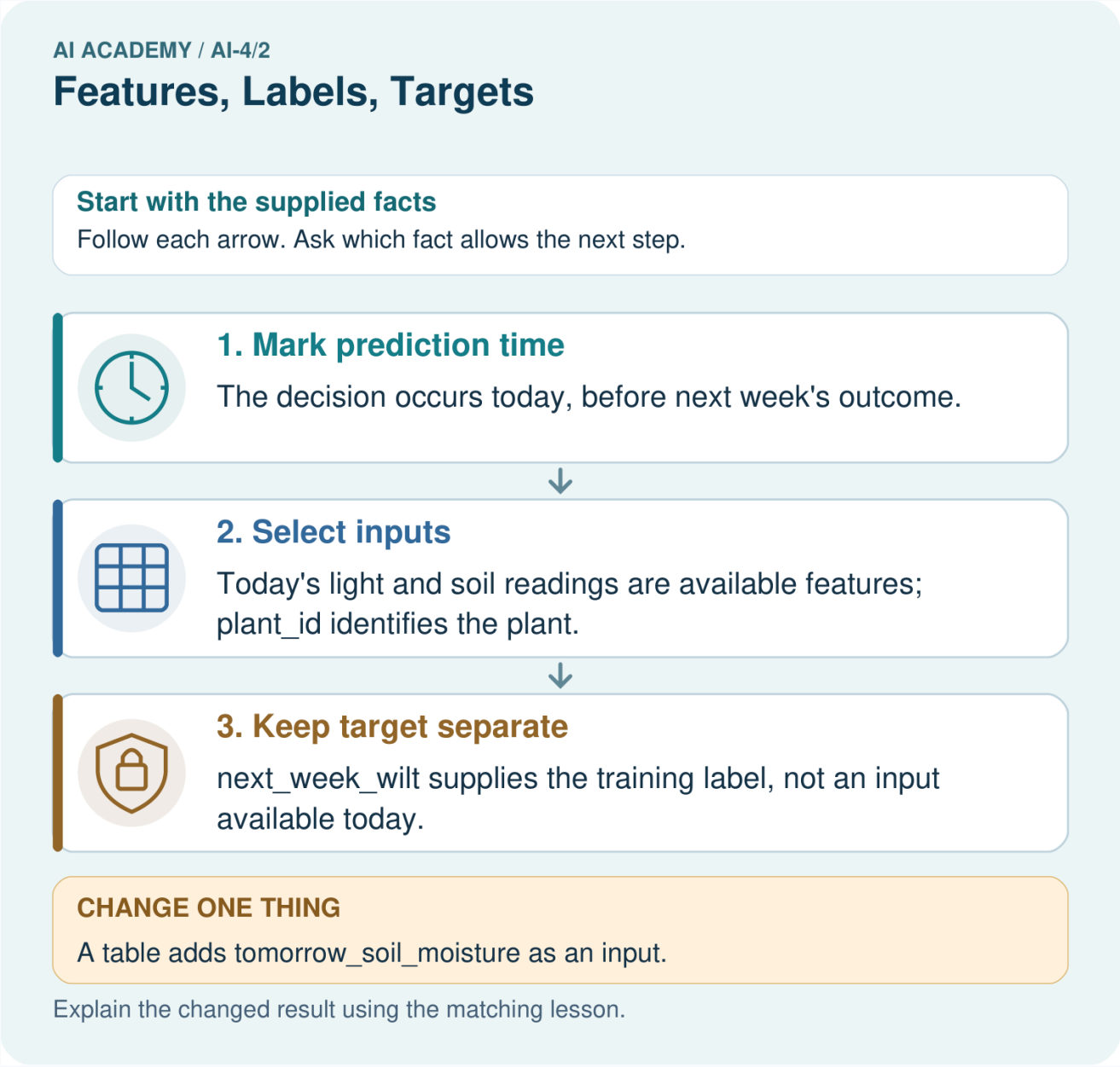
> **A completed loan has pages 200, queue_at_checkout 3 and days_to_return 14. Write the feature list and label. Someone adds overdue_fee calculated after return. Can it become a feature for the checkout prediction? Explain.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Features, Labels, Targets

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: A completed loan has pages 96, queue_at_checkout 4 and days_to_return 14. Give the ordered **feature** list and **target**. Then state what changes if it took 15 days instead.

Teacher check: The **feature** list is [96, 4]. At 14 days the **target** is 0. At 15 days the **target** becomes 1; the checkout **feature** list remains [96, 4].

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Feature**s **label**s and **prediction time**. Proposed explanation: "Every stored column is a **feature**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A historical export combines information from different times. Require a meaning and timestamp before treating a column as an **input**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short **target**ed group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Historical exports combine measurements from different times. Availability at **prediction time**, not table presence, determines eligibility. Changed-case check: Reject it for this **prediction**-time contract.

Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Changed-case reasoning: Reject it for this **prediction**-time contract. Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Rows contain leaf length, edge type and healthy/spotted **label**.

Write the question to explain: Identify **feature**s, **target** and **prediction**-time **input**s.

Reveal after **prediction**s: Length and edge type are candidate **feature**s; the **label** is the **target**. Do not supply the true **target** as an **input** to predicting it. Evaluate **feature** usefulness.

Use the earlier transfer question as additional practice. Reserve Audit the new dispatch-time schema independently for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: the ordered **input**s are [200, 3] and the observed **label** is 0 at exactly 14 days. The fee is calculated later and cannot be used for an earlier checkout **prediction**.

Exit example: use [pages, queue_at_checkout], in pages and requests, both recorded at checkout. An unknown future return outcome is not **evidence** of an on-time return and must not be silently **label**led 0.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Recall why held out labels must not choose the Week 1 threshold. Connect separate examples to the separate question of input availability.
7 to 18 minutes	State the checkout moment and label boundary. Place each field on the timeline. Explain why an arbitrary identifier can stay as metadata without becoming an input.
18 to 32 minutes	Trace the feature function on past and new records. Complete W1 and W2. Require the correct order and the 14 versus 15 day distinction.
32 to 45 minutes	Complete the guided 200-page loan and fee question. Ask which fields are unknown on a new checkout and why unknown does not mean zero.
Session 2 start to 8 minutes	Retrieve the contract: names, units, order and availability time. Ask students to explain one later field without using only the word leakage.
8 to 20 minutes	Use W3 to challenge a perfect test score. Trace how days_to_return reveals the answer even on separate test rows.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently. Require inputs, target, exclusions, boundary answer and unknown-outcome explanation before giving hints.
35 to 45 minutes	Discuss W5 and the exit contract. Use the fresh retry after support where needed and record the support given.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Library W2 features [96,4] stay fixed; target 0 at 14 days and 1 at 15. Plant inputs today's light/moisture, target next-week wilt; observed tomorrow field excluded. Puzzle inputs piece_count/clues_at_start, target slow>8 minutes; award_color is later-derived leakage. Missing or overwritten queue snapshot cannot become 0 or current export value.

Independent robot features [route_steps,queue_at_dispatch]; past [30,2] label 0 at 10 minutes, new [24,0] target unknown; trip_id metadata and later arrival/slow excluded. Require timing, not just column names.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.

2. 2. **Feature**-or-**Target** Sort: annotate three tables, explain availability time, and find one disguised post-outcome field.

3. 3. Run one normal, one boundary, one missing, and one shifted case.

4. 4. Calculate the metric from raw counts or row-level errors.

5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

Every stored column is a **feature**: A historical export combines information from different times. Require a meaning and timestamp before treating a column as an **input**.

Removing the **label** column removes leakage: days_to_return and an outcome-based fee may still disclose the **target**. Audit how every field was created.

An unknown outcome is a negative **label**: Unknown means not established. In this exercise **label**s come from completed historical returns. Do not turn unavailable outcomes into on-time **label**s.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve Audit the new dispatch-time schema independently for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Any column correlated with the answer is automatically a good **feature**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

Use [route_steps, queue_at_dispatch] in that order, measured in steps and queued trips at dispatch. slow is the **target**. Keep trip_id as tracking metadata; exclude arrival_minutes and slow from **input**s.

Past **feature**s are [30, 2] with **label** 0 because 10 is not greater than 10. New **feature**s are [24, 0]; slow is unknown until the outcome can be established.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Assign the roles

For the worked loan table, identify the two candidate **feature**s, the **target label** and the tracking field. Explain why days_to_return is not a predictor.

Candidate **feature**s are pages and queue_at_checkout.

The **target** is returned_late and the tracking field is loan_id. days_to_return is known only after return and directly defines the **target**, so it must not enter checkout **prediction**s.

W2 Trace the feature contract

A completed loan has pages 96, queue_at_checkout 4 and days_to_return 14. Give the ordered **feature** list and **target**. Then state what changes if it took 15 days instead.

The **feature** list is [96, 4]. At 14 days the **target** is 0. At 15 days the **target** becomes 1; the checkout **feature** list remains [96, 4].

W3 Repair a misleading score

A model has every test **prediction** correct after using pages and days_to_return. A learner says a separate test set makes this valid. Explain the flaw and repair the **input**s.

The test rows contain future information that reveals the **label**. Separate rows do not make days_to_return available at checkout.

Remove later outcome fields, use the checkout **feature contract**, and rerun development and evaluation with honest records.

W5 Handle an unavailable input

For the robot task, a dataset stores queue_now at export time but has no saved queue_at_dispatch. Can you rename that field to make it usable? State a defensible next step.

Renaming does not restore the missing dispatch-time value. Recover a reliable timestamped dispatch snapshot if available, or remove the **feature** and reevaluate the revised design.

Do not invent a zero queue or copy the later count.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Length and edge type are candidate **feature**s; the **label** is the **target**. Do not supply the true **target** as an **input** to predicting it. Evaluate **feature** usefulness.

Additional transfer practice question

A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two **feature**s and the **target**.

Reasoned practice answer

Leaf length and leaf width are **feature**s; species is the **target label**. Species must not also be used as an **input** for this **prediction** task.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The **target** can be included as an **input** because it improves apparent accuracy.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Verified species is the **target** and must not be supplied as a **prediction feature**. Use length and edge type if available at **prediction time**, then evaluate their usefulness.

Numeric encoding of the **target** does not make it a legitimate **input**.

Explanations for the additional workbook examples

W1 Assign the roles

W2 Trace the feature contract

W3 Repair a misleading score

W4 Audit a new task independently

A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

Use [route_steps, queue_at_dispatch] in that order, measured in steps and queued trips at dispatch. slow is the **target**. Keep trip_id as tracking metadata; exclude arrival_minutes and slow from **input**s.

Past **feature**s are [30, 2] with **label** 0 because 10 is not greater than 10. New **feature**s are [24, 0]; slow is unknown until the outcome can be established.

W5 Handle an unavailable input

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Define the Plant Classifier **target** and a first **feature** list; record units, allowed values, collection method, and availability time.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny **label**led dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Place field cards before or after the **prediction**-time line. Supply an empty contract table, not completed answers. Extension: a queue measurement is recorded before checkout but overwritten at export.

Explain why its current value is not necessarily the original snapshot, and propose retaining timestamped history.

Require W4 **feature**s [30, 2], **target** 0, new **feature**s [24, 0], unknown new **target**, and the timing reason for exclusions. A column list without timing justification is insufficient.

Fresh retry: at print-job submission, predict whether completion takes more than 5 minutes. **Input**s are page_count then queue_at_submit. A completed job has 12 pages, queue 1, completion_minutes 6: **input**s [12, 1], **label** 1.

A new job has 8 pages, queue 0: **input**s [8, 0], **label** unknown. Exclude completion_minutes and the **label** from **input**s; keep job_id as metadata. Exactly 5 minutes would have **label** 0.

Delayed return

Suggested return points in this level: ai-4/3, ai-4/5, ai-4/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 3: Classification or Regression?. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 3 distinguishes classification **target**s from regression **target**s. Carry forward the explicit **target** definition and **prediction time**.

Teacher reference: https://scikit-learn.org/stable/common_pitfalls.html#data-leakage (reviewed 2026-09-26).

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching **target**: Separate **prediction input**s from observed **target**s

Prepare the **label**led worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: A completed loan has pages 96, queue_at_checkout 4 and days_to_return 14. Give the ordered **feature** list and **target**. Then state what changes if it took 15 days instead.

Reasoned practice answer: The **feature** list is [96, 4]. At 14 days the **target** is 0. At 15 days the **target** becomes 1; the checkout **feature** list remains [96, 4].

Check every learner before discussion: Use the worked case for **Feature**s **label**s and **prediction time**. Proposed explanation: “Every stored column is a **feature**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A historical export combines information from different times. Require a meaning and timestamp before treating a column as an **input**.

Support: read the exact case aloud, enlarge the **label**led picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A table adds tomorrow_soil_moisture as an **input**. Require a **prediction** and a reason before comparing. Expected result: Reject it for this **prediction**-time contract.

Reason: Tomorrow’s measurement is unavailable when today’s **prediction** must be issued.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

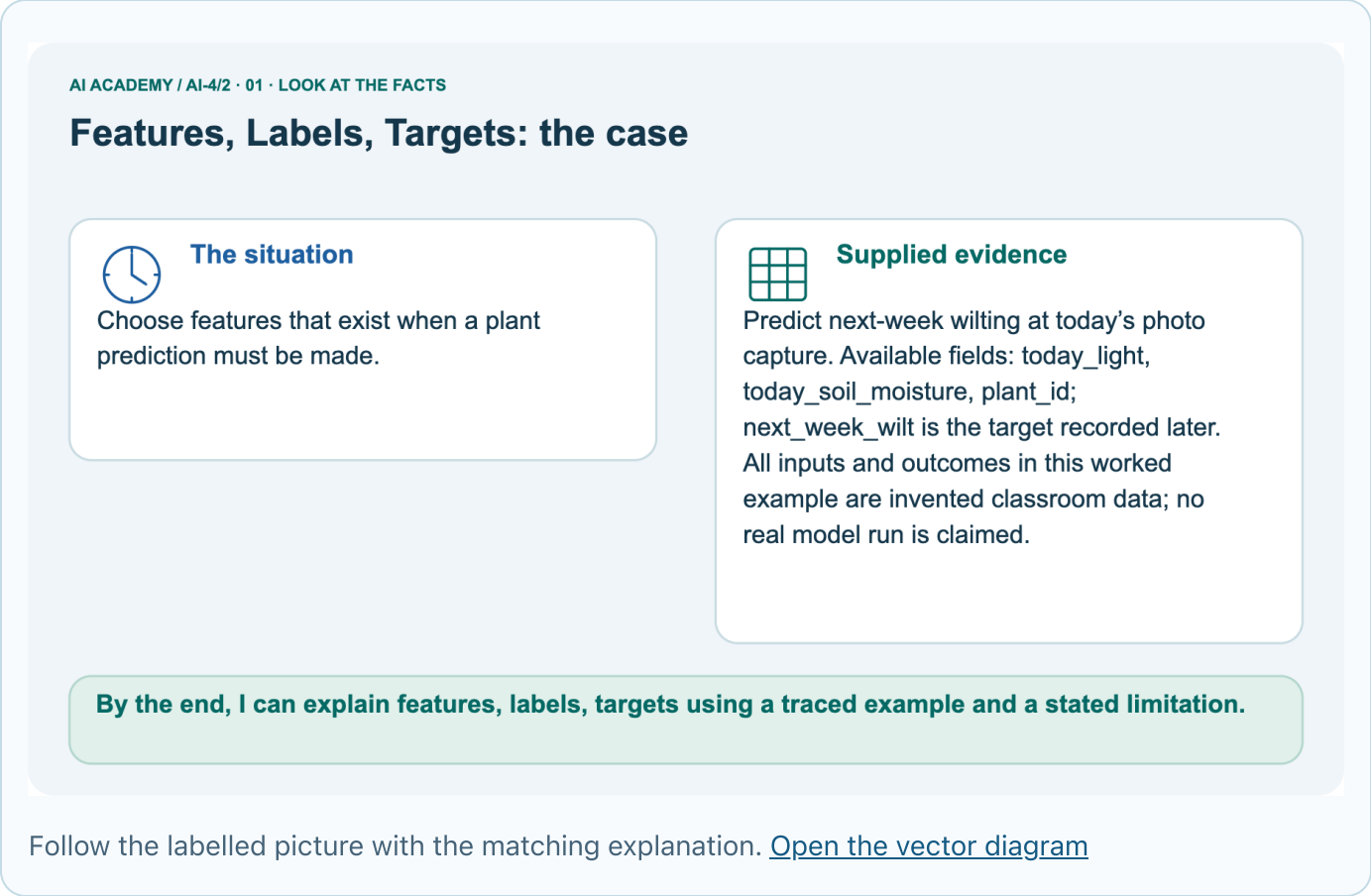
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Features, Labels, Targets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

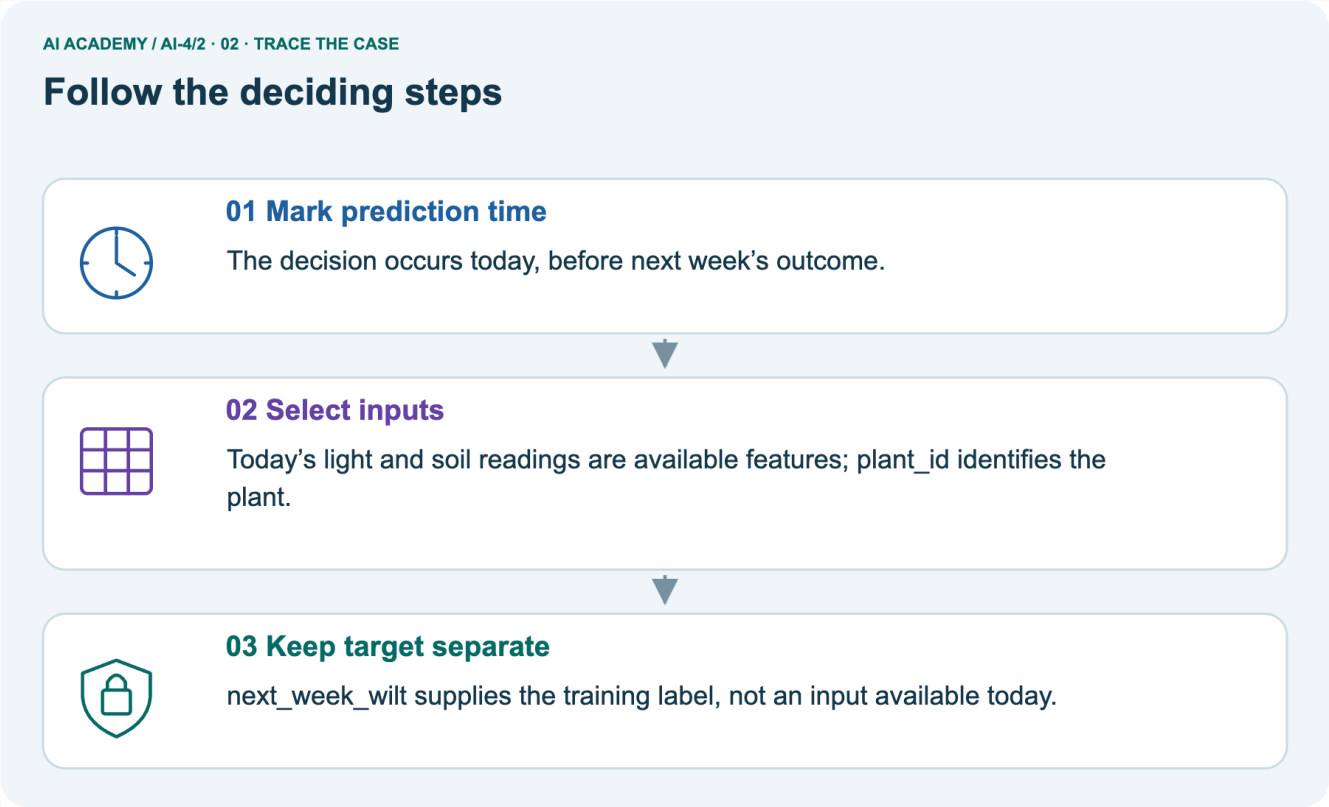
Choose **feature**s that exist when a plant **prediction** must be made. Supplied facts: Predict next-week wilting at today's photo capture.

Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



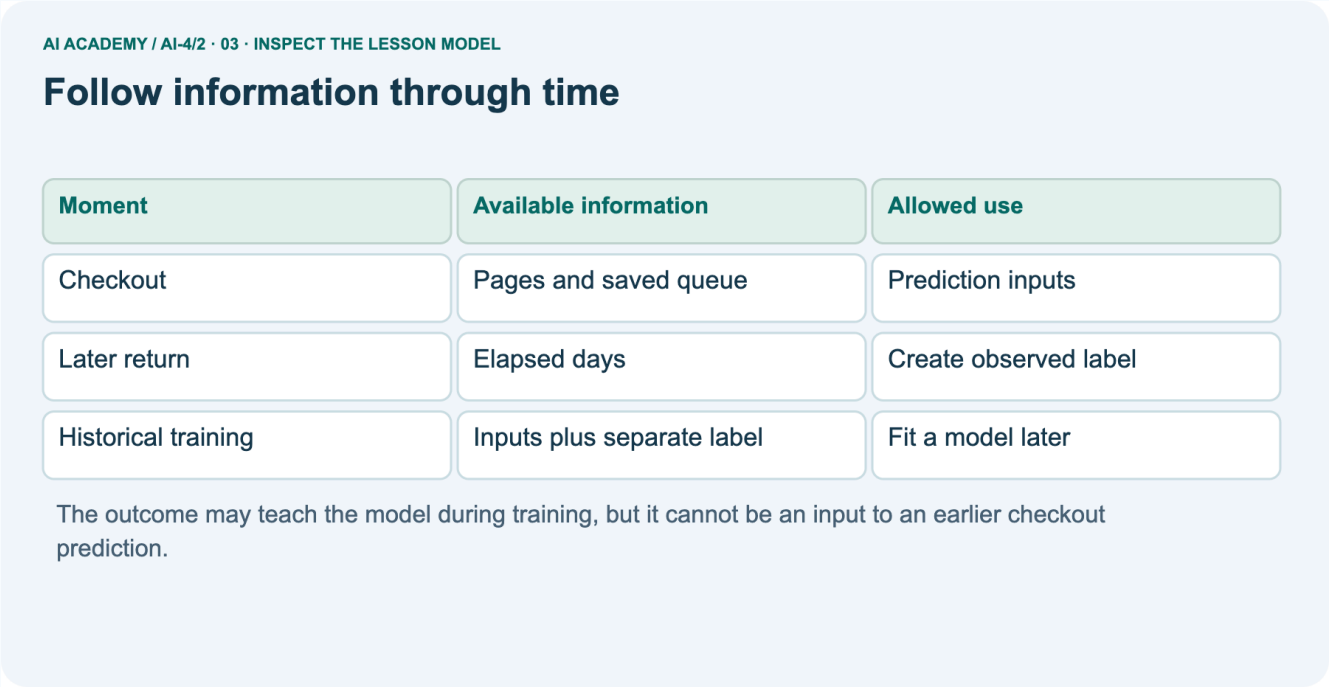
Follow the labelled picture with the matching explanation. Open the vector diagram

Mark **prediction time**: The decision occurs today, before next week's outcome. → Select **input**s: Today's light and soil readings are available **feature**s; plant_id identifies the plant. → Keep **target** separate: next_week_wilt supplies the **training label**, not an **input** available today.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow information through time



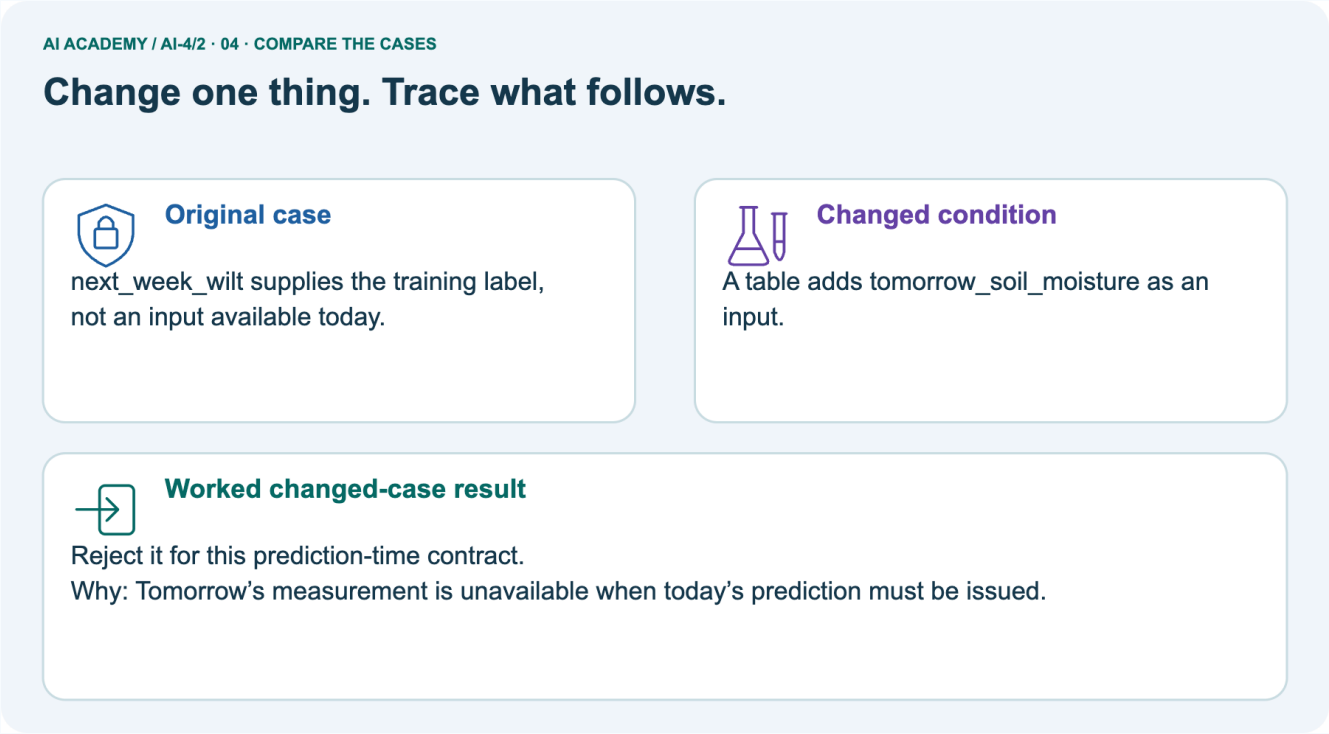
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow information through time. Checkout | Pages and saved queue | **Prediction input**s. Later return | Elapsed days | Create observed **label**. Historical **training** | **Input**s plus separate **label** | Fit a model later

Read the **label**s and values. Explain the relationship the model shows.

Teacher prompt: Read the **label**s and values. Explain the relationship the model shows. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: next_week_wilt supplies the **training label**, not an **input** available today. Changed condition: A table adds tomorrow_soil_moisture as an **input**. Expected result: Reject it for this **prediction**-time contract.

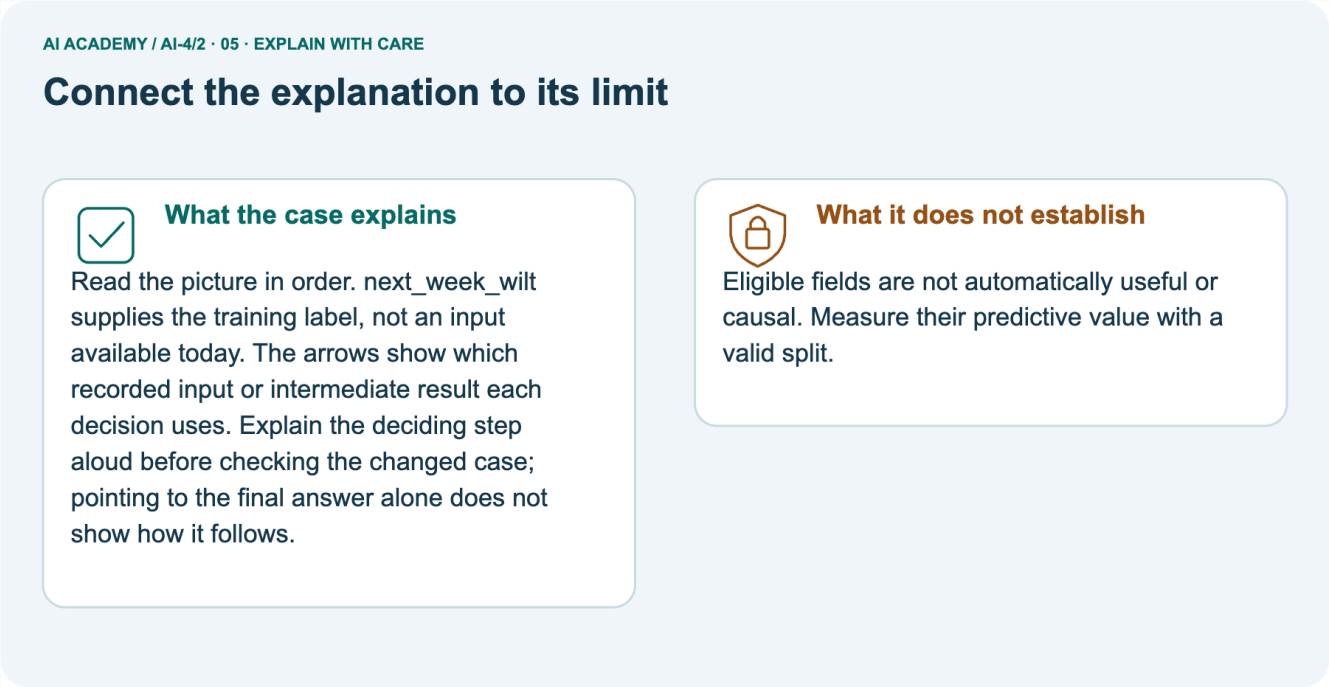
Reason: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one **label** or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Eligible fields are not automatically useful or causal.

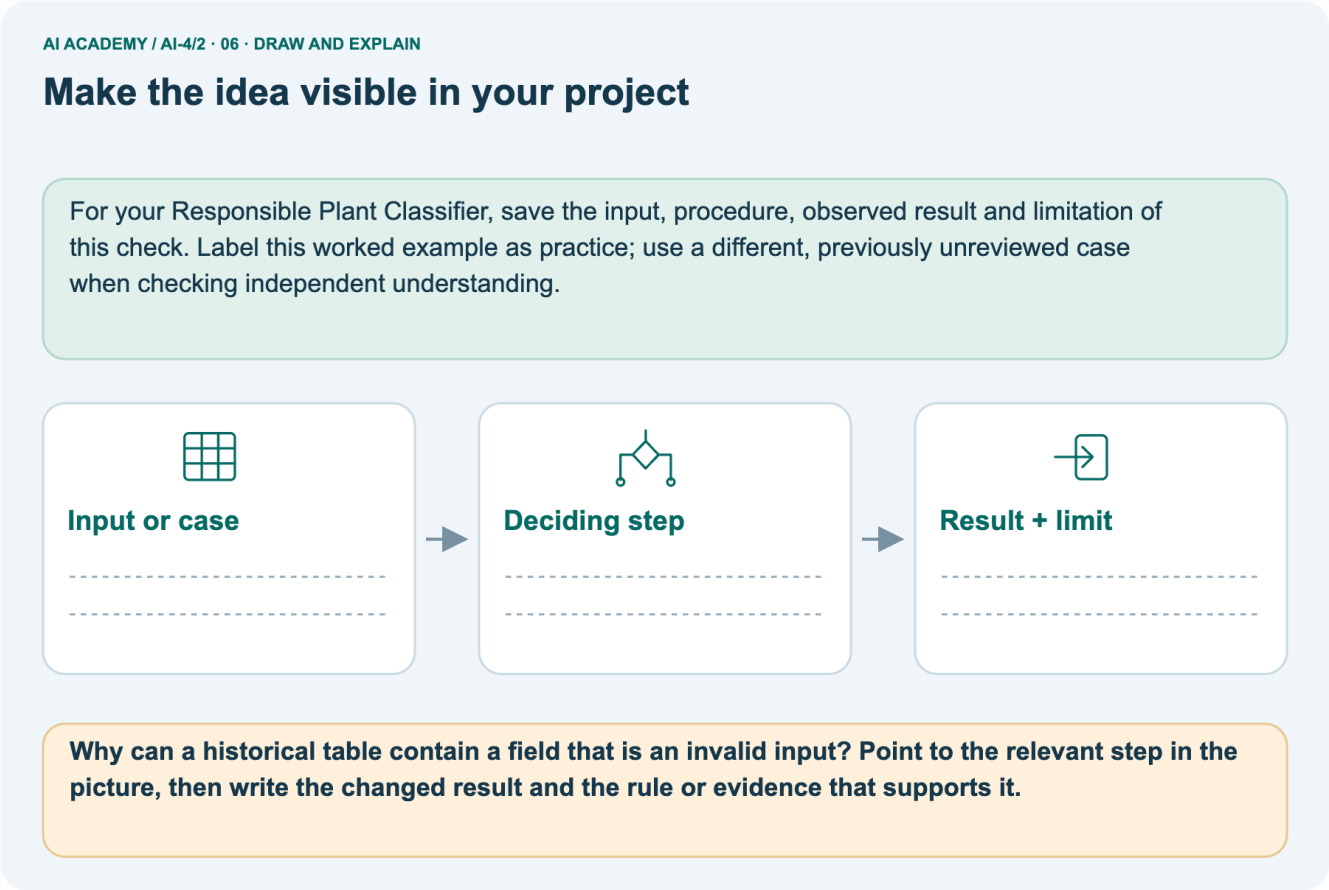
Measure their predictive value with a valid split.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Eligible fields are not automatically useful or causal. Measure their predictive value with a valid split.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Historical exports combine measurements from different times. Availability at **prediction time**, not table presence, determines eligibility.

Changed-case check: Reject it for this **prediction**-time contract. Tomorrow's measurement is unavailable when today's **prediction** must be issued.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Features, Labels, Targets: the case

02. Follow the deciding steps

03. Follow information through time

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.